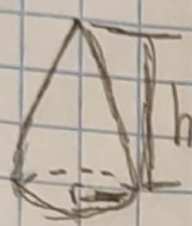
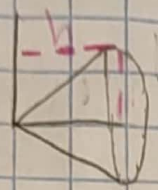
tarea sección 6.2a

479
1309

47) Un cono circular cuya altura es h y el radio de la base es r



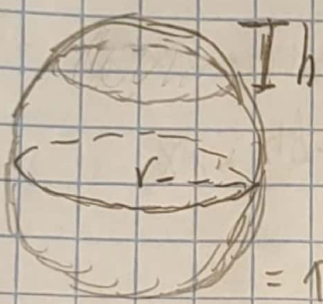


$y = \frac{r}{h} x$ $\quad V = \pi \int_a^b f(x)^2 \, dx$

$V = \int_0^h \frac{r^2}{h^2} x^2 \, dx = \pi \frac{r^2}{h^2} \int_0^h x^2 \, dx$

$= \pi \frac{r^2}{h^2} \cdot \frac{x^3}{3} \Big|_0^h = \frac{\pi r^2 h^3}{3h^2} = \frac{1}{3} \pi r^2 h$

49)

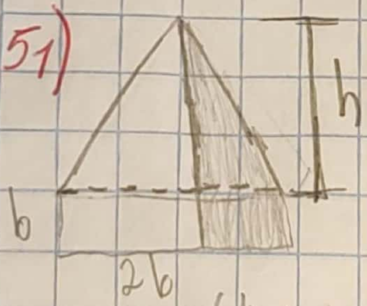


$V = \pi r^2 h \quad y^2 = r^2 - x^2 \quad V = \frac{\pi h^2}{3}(3r - h)$

$V = \pi \int_a^b f(x)^2 \, dx = \pi \int_0^h (\sqrt{2xr - x^2})^2 \, dx$

$= \pi \int_0^h 2xr - x^2 \, dx = \pi \cdot \frac{2x^2}{2} r - \frac{x^3}{3} \Big|_0^h = \pi h^2 \left(r - \frac{h}{3}\right)$

51)



$V = (b \cdot 2b) \, dx$

$\frac{x}{b} = \frac{h-y}{h} = x = \frac{b}{h}(h-y) = \frac{2b^2}{h^2}(h-y)^2$

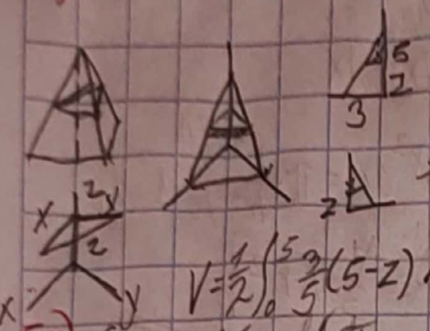
$V = \frac{2a^2}{h^2} \int_0^h (h^2 - 2hy + y^2) \, dx = \frac{2b^2}{h^2}\left(h^2 y - h y^2 + \frac{1}{3} y^3\right)$

$= \frac{b^2}{h^2}\left(\frac{1}{3} h^3\right) - \frac{1}{3} h (2b^2)$

$V = \int_0^h 2b^2 \left(\frac{x^2}{h^2}\right) dx = 2 \frac{b^2}{h^2} \left(\frac{x^3}{3}\right) \Big|_0^h$

$2b^2 \left(\frac{1}{3} h\right)$

59) Un tetraedro con 3 caras mutuamente perpendiculares 3 aristas



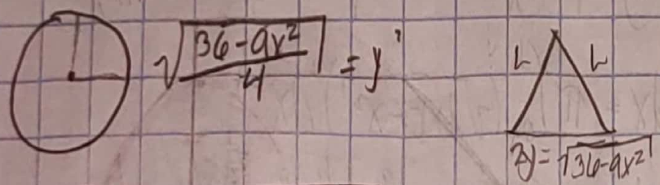
$\frac{5}{3} = \frac{5-z}{x}$ $\quad x = \frac{3}{5}(5-z)$ $\quad V = \frac{1}{2} xy \, dz$

$\frac{5}{4} = \frac{5-z}{y}$ $\quad y = \frac{4}{5}(5-z)$ $\quad V = \int_0^5 \frac{1}{2}\left[\frac{3}{5}(5-z)\right]\left[\frac{9}{5}(5-z)\right] dz$

$V = \frac{1}{2}\int_0^5 \frac{3}{5}(5-z) \cdot \frac{4}{5}(5-z)\, dz = \frac{1}{2}\left[\frac{3}{5}\int_0^5 5-z + \frac{4}{5}\int_0^5 5-z\right] = 10\ cm^3$

55) S = región elíptica $\quad 9x^2 + 4y^2 = 36$

triángulos rectángulos isósceles



$\sqrt{\frac{36-9x^2}{4}} = y$

$2y = \sqrt{36-9x^2}$

$2L^2 = H^2 = L^2 = \frac{36-9x^2}{2}$

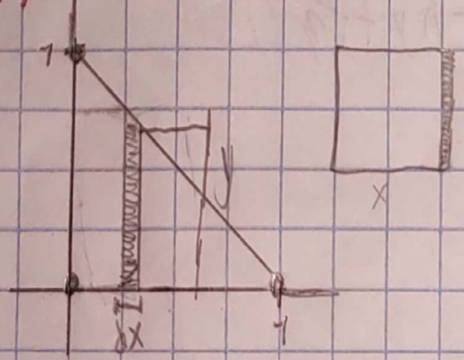
$A(x) = \frac{L^2}{2} = A(x) = \frac{36-9x^2}{4}$

$\sqrt{\frac{36-9x^2}{2}} = y_1 = \pm\sqrt{\frac{36-9x^2}{2}}$

$V = \int_{-2}^{2} A(x)\, dx = \int_{-2}^{2} \frac{36-9x^2}{4}\, dx = 2\int_0^2 \left[\frac{36-9x^2}{4}\, dx = \int_0^2 18 - \frac{9}{2}x^2\, dx\right.$

$18x - \frac{9}{2}\frac{x^3}{3}\Big|_0^2 = 18(2) - \frac{9}{2}\frac{2^3}{3} = 24\ u^3$

57)



$V = a \cdot g$

$V = \int^2 dx$

$V = \int_0^1 x^2\, dx$

$V = \frac{x^3}{3}\Big|_0^1 = \frac{1^3}{3} = \frac{1}{3}\ u^3$

$m = \frac{0-1}{1-0} = \frac{-1}{1} = -1$ $\quad V = y^2 dx$

$V = \int_0^1 (-x+1)^2\, dx$

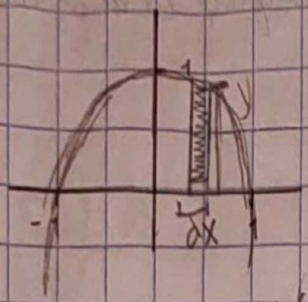
$y - 0 = -1(x-1)$

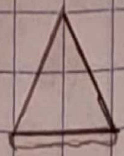
$y = -x + 1$

59) $y = 1 - x^2$ y el eje x

$V = a \cdot g$

$V = \left(\frac{1}{2} bh\right) dx$





$V = \frac{1}{2}(1-x^2)(1-x^2)\,dx$

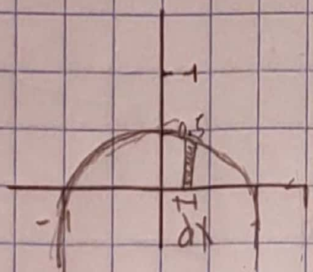
$V = \frac{1}{2}\int_{-1}^{1} x^4 - 2x^2 + 1\,dx$

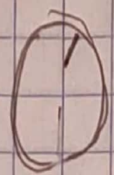
$V = \frac{1}{2}\left(\frac{x^5}{5} - \frac{2x^3}{3} + x\right)\Big|_{-1}^{1} = \frac{4}{15} - \left(-\frac{4}{15}\right) = \frac{8}{15}\,u^3$

61) $y = \frac{1}{2}(1-x^2) \qquad -1 \leq x \leq 1$

$V = a \cdot g$

$V = (\pi r^2)\,dx$





$V = \pi\left(\frac{1}{2}(1-x^2)\right)^2 dx$

$V = \pi\int_{-1}^{1} \frac{1}{4} - \frac{1}{2}x^2 + \frac{1}{4}x^4\,dx$

$V = -\frac{\pi}{4}\left(\frac{1}{3}x^3 + \frac{1}{5}x^5\right)\Big|_{-1}^{1}$

$V = \frac{2}{15}\pi - \left(-\frac{2}{15}\pi\right) = \frac{4}{15}\pi$

63) a) $(x-0)^2 + (y-R)^2 = r^2$

$(y-R)^2 = r^2 - x^2 = R = \pm\sqrt{r^2 - x^2}$

$V = \int_{-r}^{r} \pi f(x)^2 - g(x)^2\,dx =$

$= \pi\int_{-r}^{r}\left[\left(R + \sqrt{r^2-x^2}\right)^2 - \left(r - \sqrt{r^2-x^2}\right)^2\right]dx$

$V = 2\pi\int_{-r}^{r} 4R\sqrt{r^2 - x^2}\,dx = 8\pi R\int_0^r \sqrt{r^2-x^2}\,dx$

b) $2\pi\int_0^r 4R\sqrt{r^2-x^2}\,dx = 8\pi r\int_0^{\pi/2}\sqrt{r^2\cos^2\theta}\,r\,d\theta$

$= 8\pi R r^2\int_0^{\pi/2}\cos^2\theta\,d\theta = 8\pi R r^2\int_0^{\pi/2}\frac{1}{2}(1+\cos 2\theta)\,d\theta$

$= 4\pi R r^2\left(\theta + \frac{\operatorname{sen} 2\theta}{2}\right)\Big|_0^{\pi/2} = 4\pi R r^2\,\frac{\pi}{2} = 2\pi^2 r^2 R$

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